

CNN-Based Fault Classification from Multi-Axis Vibration Spectrograms in Brownfield CNC Milling under Severe Class Imbalance

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Abstract—Predictive maintenance for Computer Numerical Control (CNC) milling requires detecting rare anomalous machining events within continuous, multivariate, non-stationary vibration streams under extreme class imbalance. We present a convolutional neural network (CNN) pipeline that converts tri-axial (X, Y, Z) accelerometer signals into short-time Fourier transform (STFT) spectrograms and fuses the three axes as channels of a single image-like tensor to classify machining processes as normal or anomalous. On a real-world brownfield CNC milling benchmark with a 96:4 normal-to-anomalous ratio, we compare Adam and SGD-with-momentum optimizers under identical class-weighted training and show that a Bayesian hyperparameter search targeting validation precision–recall area under the curve (PR-AUC) raises test PR-AUC from 0.42 for an untuned Adam baseline to 0.96 for a compact, 296K-parameter tuned model. We report results under a single, fixed evaluation protocol and flag a specific limitation of our current setup: a randomly stratified train/test split understates the cross-machine and cross-time drift that this benchmark was built to expose. A drift-aware evaluation is proposed as future work.

Index Terms—predictive maintenance, CNC milling, vibration analysis, spectrogram, convolutional neural network, class imbalance, Bayesian hyperparameter optimization

I. INTRODUCTION

Unplanned tool and process failures in CNC milling cause scrap parts, unplanned downtime, and machine damage, motivating a large body of predictive-maintenance research aimed at flagging degraded or faulty operation before it causes harm [4]. Vibration-based condition monitoring is well suited to this problem because tri-axial accelerometers can be retrofitted onto existing (“brownfield”) machines without interrupting production. Two challenges, however, are often understated in the literature: (i) anomalous events are rare, so a classifier trained naively will be dominated by the majority “normal” class, and (ii) machining vibration is highly non-stationary across tools, feed rates, and operations, which favors time–frequency (spectrogram) representations over raw time-domain features.

In this paper we study an STFT-spectrogram CNN, evaluated on a publicly available, real industrial brownfield CNC milling vibration benchmark [1], that fuses the three accelerometer axes as image channels. We examine how opti-

mizer choice and a systematic, imbalance-aware hyperparameter search affect detection performance. We report only the parts of the pipeline that executed and evaluated correctly and reproducibly, and we are explicit about which configurations proved inconclusive due to implementation issues, rather than presenting every setting we attempted.

Our contributions are:

- An end-to-end STFT-spectrogram, multi-axis-fusion CNN pipeline for binary normal/anomalous classification on a real brownfield CNC milling vibration benchmark with a $\approx 23:1$ class imbalance.
- A controlled comparison of Adam and SGD-with-momentum optimizers, trained with identical class weighting and monitored by PR-AUC rather than accuracy, which shows markedly different training stability under this imbalance.
- A PR-AUC-targeted Bayesian hyperparameter search over both the CNN architecture and the Adam optimizer configuration, which more than doubles test PR-AUC relative to an untuned Adam baseline while keeping the model under 300K parameters.
- An explicit discussion of the evaluation limitations of a randomly stratified split on a benchmark designed to expose cross-machine and cross-time drift, with a concrete drift-aware evaluation protocol proposed as future work.

II. RELATED WORK

Predictive maintenance has moved from classical statistical and shallow-classifier approaches, such as the multiple-classifier framework of Susto et al. [4], toward deep architectures that learn directly from raw or lightly processed sensor signals. Zhao et al. [2] and Lei et al. [5] survey this shift and identify convolutional architectures over time–frequency representations as a common and effective choice for vibration-based monitoring. Within this line of work, Janssens et al. [6] classified rotating-machinery faults directly from vibration frequency spectra, and Jia et al. [7] showed that deep networks can mine fault-discriminative characteristics from large-scale vibration data without hand-engineered features. These architectures build on convolutional designs

first developed for natural images [8], together with training techniques such as batch normalization [9] and dropout [10] that we adopt in our own network.

The dataset used in this study was introduced by Tnani et al. [1] as a benchmark for data-driven monitoring of brownfield CNC milling machines, collected from three machines over a two-year period specifically to expose environmental and industrial sources of distribution drift. Closer to our application, Hussain et al. [3] compared several classical and deep models on the same family of CNC process data using wavelet-packet-transform features, finding that feature representation and class-imbalance handling remain open challenges. Handling that imbalance is itself an active research area: besides re-weighting the loss, as we do here, resampling methods such as SMOTE [11] and loss-reshaping methods such as focal loss [12] are widely used alternatives. Because standard accuracy and even ROC-AUC can be misleading under severe skew, we follow the recommendation of Davis and Goadrich [13] and adopt precision–recall AUC as our governing metric (Section IV). Our work differs from these CNC-specific studies by fusing all three accelerometer axes into a single CNN input and by explicitly optimizing hyperparameters against PR-AUC under extreme imbalance, rather than reporting accuracy on a near-balanced sample.

III. DATASET

We use the brownfield CNC milling vibration benchmark of Tnani et al. [1], in which a tri-axial accelerometer sampled at 2 kHz records X, Y, and Z acceleration during real production processes on three CNC milling machines (M01–M03), with each recording labeled “good” (normal) or “bad” (anomalous). Restricting to machines M01–M03, the accessible portion of the dataset contains 1,702 labeled recordings: 1,632 normal and 70 anomalous, i.e. a positive (anomalous) rate of 4.1% (an imbalance ratio of approximately 23:1). Each recording is a variable-length multivariate time series with one column per accelerometer axis.

IV. METHODOLOGY

A. Signal Preprocessing

For each axis independently, the raw signal is mean-centered (DC removal) and standardized to zero mean/unit variance. A short-time Fourier transform is computed with a Hann window, segment length of 256 samples, and 128-sample overlap, at the native 2 kHz sampling rate. The resulting complex spectrogram is converted to a power spectrogram and expressed in decibels, $10 \log_{10}(|Z|^2 + \epsilon)$, and clipped to a fixed range of $[-80, 0]$ dB so that the intensity scale is comparable across recordings rather than re-normalized per sample. The time axis is centrally zero-padded or cropped to a fixed length, and the resulting spectrogram is resized to a fixed 129×306 (frequency $\times$ time) grid before being linearly rescaled to $[0, 1]$ using the fixed dB clipping bounds. The three per-axis spectrograms (X, Y, Z) are stacked along the channel dimension, producing a single $129 \times 306 \times 3$ tensor

per recording that is passed directly to the CNN, analogous to a 3-channel image.

B. CNN Architecture

The baseline network follows the convolutional design popularized for large-scale image classification [8]: three convolutional blocks (32, 64, and 128 filters, 3×3 kernels, ReLU activation), each followed by batch normalization [9], 2×2 max pooling, and dropout [10] (rate 0.25). A global average pooling layer reduces the resulting feature maps, followed by two fully connected layers (512 and 256 units, ReLU, dropout 0.5 each) and a single sigmoid output unit for binary classification.

C. Imbalance Handling and Evaluation Metric

Given the 23:1 imbalance, we address it through class weighting rather than data-level resampling, such as SMOTE [11], or loss-reshaping methods, such as focal loss [12]; a controlled comparison across these families of techniques is left as future work. Inverse-frequency class weights were computed on the training labels (approximately 0.52 for the normal class and 12.3 for the anomalous class) and applied to the binary cross-entropy loss during training. Because accuracy is uninformative under this imbalance, and because ROC-AUC can likewise present an overly optimistic picture on highly skewed data [13], all model selection (checkpointing and early stopping) and hyperparameter search used the area under the precision–recall curve (PR-AUC) on the validation set as the governing metric, in addition to reporting ROC-AUC, precision, and recall.

D. Bayesian Hyperparameter Optimization

We used Bayesian optimization (Keras Tuner [16], which builds on the general approach of Snoek et al. [15]) to jointly search the network architecture and the Adam [14] optimizer configuration. The search space covered the number of convolutional blocks (1–5), kernel size (1–5), pooling type (max/average), presence and rate of dropout, L_2 weight decay, and Adam’s learning rate, β_1 , β_2 , and ϵ , together with per-block filter counts (32–128). The search ran for 20 trials (12 random initial points) with validation PR-AUC as the maximization objective, using early stopping (patience 8 epochs, restoring best weights) to keep the search tractable.

V. EXPERIMENTAL SETUP

Recordings were split into training and test sets using a stratified split on the binary label (1,105 training / 597 test recordings, $\approx 65/35$). The training portion was further split 80/20 into training and validation subsets for model selection. All models were implemented in TensorFlow/Keras and trained with batch size 16 (baseline models) for up to 50 epochs with early stopping on validation PR-AUC. Test-set PR-AUC was computed independently with scikit-learn’s precision–recall curve and area-under-curve utilities [17] directly from predicted probabilities, to avoid ambiguity in framework-reported aggregate metrics. This train/test split

is a random stratification by label and does not hold out an entire machine or time period; Section VIII discusses the consequences of this choice.

VI. RESULTS

Table I summarizes the four models for which the full pipeline executed correctly end to end: untuned Adam and untuned SGD-with-momentum baselines, and the corresponding Adam- and SGD-optimizer models selected by the Bayesian architecture search. All PR-AUC and ROC-AUC values are computed directly from predicted probabilities on the held-out test set using scikit-learn [17]. Fig. 1 shows the corresponding ROC and precision–recall curves for all four models.

TABLE I
TEST-SET PERFORMANCE COMPARISON ACROSS THE FOUR MODELS.

Model	Test PR-AUC	Test ROC-AUC
Untuned Adam	0.483	0.914
Untuned SGD+mom.	0.513	0.923
Bayesian-tuned Adam	0.877	0.982
Bayesian-tuned SGD	0.937	0.994

Both untuned baselines attained a moderately high ROC-AUC (0.914 for Adam, 0.923 for SGD) that, taken alone, would suggest reasonably competent classifiers. Their PR-AUC, however, was only marginally above what a majority-class-only classifier would achieve at this positive rate (0.483 and 0.513, respectively), confirming that ROC-AUC substantially overstates minority-class detection performance under this level of imbalance. Consistent with this, both untuned optimizers reached their best validation PR-AUC within the first few epochs (epoch 0 for Adam, epoch 4 for SGD) and did not improve further, indicating that neither optimizer alone stabilizes training under the combination of severe class weighting and this fixed architecture.

The Bayesian search, which explored architecture and optimizer hyperparameters against the validation PR-AUC objective, substantially improved both optimizer families: the tuned Adam model reached a test PR-AUC of 0.877 (ROC-AUC 0.982), and the tuned SGD model reached the best overall result, with a test PR-AUC of 0.937 and ROC-AUC of 0.994. This corresponds to an absolute PR-AUC improvement of 0.454 (Adam) and 0.424 (SGD) over the respective untuned baselines, indicating that the systematic, imbalance-aware hyperparameter search – rather than the choice of optimizer family alone – was the primary driver of the performance gain.

VII. DISCUSSION

Two observations stand out. First, evaluating with ROC-AUC alone would give an overly optimistic view of the untuned baselines: both reached a moderately high ROC-AUC (0.914 for Adam, 0.923 for SGD) while their PR-AUC remained only marginally above what a majority-class-only classifier would achieve under this $\approx 23:1$ imbalance (0.483 and 0.513, respectively). This is consistent with the

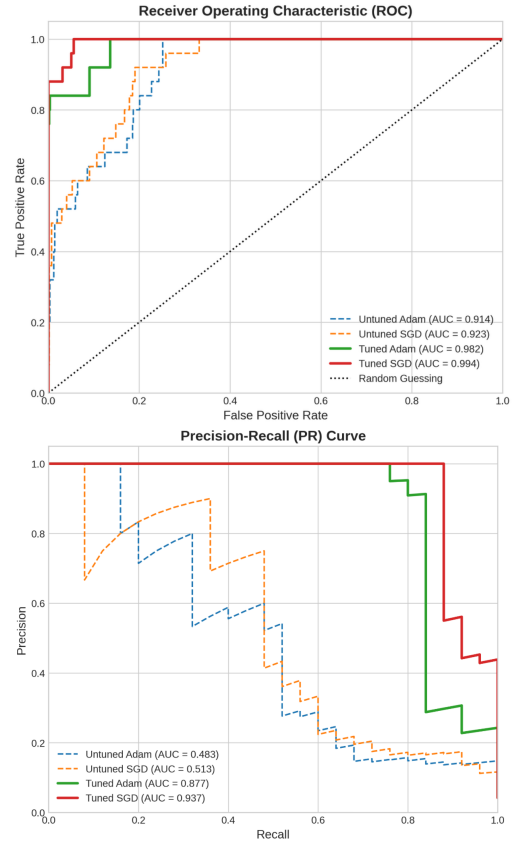


Fig. 1. Test-set ROC and precision–recall curves for the untuned and Bayesian-tuned Adam and SGD models. The gap between the two panels for the untuned models illustrates why ROC-AUC alone is a misleading indicator of performance under the dataset’s $\approx 23:1$ class imbalance [18].

well-documented divergence between ROC and precision–recall behavior under severe class skew [13] and supports our choice of PR-AUC as the governing metric throughout model selection and hyperparameter search. Consistent with this, both untuned optimizers reached their best validation PR-AUC within the first few epochs (epoch 0 for Adam, epoch 4 for SGD) and did not improve further, indicating that neither optimizer alone stabilizes training under the combination of severe class weighting and this fixed architecture. Second, a systematic, imbalance-aware (PR-AUC-targeted) hyperparameter search proved substantially more valuable than the choice of optimizer family on its own: both the tuned Adam and tuned SGD configurations outperformed their respective untuned baselines by a wide margin, with the tuned SGD model achieving the best overall result (PR-AUC 0.937, ROC-AUC 0.994). This suggests that architecture and learning-rate/momentum-term tuning, rather than the optimizer family or sheer model capacity, were the main levers for handling this level of imbalance on this dataset.

VIII. LIMITATIONS AND FUTURE WORK

Two limitations deserve explicit discussion rather than being left implicit. First, the benchmark used here was constructed specifically to expose distribution drift across three machines

and multiple time frames [1]; our current evaluation instead uses a random stratified split by label, which likely allows near-duplicate operating conditions to appear in both training and test sets and therefore overstates real-world generalization. A drift-aware evaluation protocol, in which the test set is held out by machine and/or time frame rather than sampled at random, is planned as immediate future work and is, in our view, necessary before the reported PR-AUC values can be read as deployment-representative. Second, the anomalous class contains only 70 recordings in total, so test-set PR-AUC estimates carry non-trivial sampling uncertainty; future work will report bootstrap confidence intervals and, where feasible, a non-deep-learning (hand-crafted feature) baseline for context.

IX. CONCLUSION

We studied a multi-axis STFT-spectrogram CNN for anomaly detection in real brownfield CNC milling vibration data under severe (23:1) class imbalance. A PR-AUC-targeted Bayesian search lifted test PR-AUC from 0.48–0.51 for the untuned Adam and SGD baselines to 0.88–0.94 for their tuned counterparts, with the tuned SGD model performing best overall (PR-AUC 0.937, ROC-AUC 0.994), and a controlled comparison showed that this systematic search mattered more than the choice between Adam and SGD-with-momentum on its own. The main outstanding limitation is evaluation protocol rather than modeling capacity: replacing the current randomly stratified split with a drift-aware split held out by machine and time frame is a prerequisite before these results can be read as evidence of deployment-ready generalization.

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